

Data Visualization Analysis on a Systematic Literature Review of Twitter as a Predictive System

1. Introduction

The paper 'Twitter as a predictive system: A systematic literature review' written by Cano-Marín, Mora-Cantalops and Sánchez-Alonso (2023) is selected for evaluating and analyzing the use of visuals in academic research. The study aimed to address the gap in systematic literature reviews (SLRs) on Twitter's predictive capacity by examining both research and business applications. The analysis focuses on how the authors utilized visual elements to present their findings and communicate complex information effectively.

The literature review was conducted following Kitchenham's (2004) recommendations and guidelines. A total of 7,928 documents and publications related to Twitter, dating from 2007 to November 2021, were collected from multiple data sources including ScienceDirect, Scopus, Web of Science, IEEE, Google Scholar, and Dimensions.ai. The search queries focused on Twitter-related predictive or forecasting studies, excluding duplicates, invalid DOIs, and non-English publications.

The final selection of 3,250 articles was tabulated to show research results, including their title, abstract, publication year, application domain, journal impact factor, and citation information regarding Twitter's use as a predictive system. These publications were manually classified by field of research (application domain) based on title, abstract, and content when necessary. Each published article was assigned only one predominant application domain, following categories mentioned by Kursuncu (2019).

Furthermore, the literature review analysis was enhanced using machine learning and graph analytics, specifically topic modelling, allowing for the processing of a larger number of articles compared to traditional SLRs. The review confirms Twitter's potential as a predictive system, demonstrating increased research activity and diverse applications across multiple fields. The research findings were visually presented using various charts and data tables.

The selected paper contains 13 visual items: 3 simple charts and 11 data tables detailing the statistical results of the SLRs analysis. In my data visualization analysis, I have examined all graphical presentations employed throughout the paper to understand the research work and how the authors presented a comprehensive view of Twitter's potential as a predictive system for researchers and business leaders.

In my visualization report, I begin by analyzing individual charts and tables, providing feedback based on data visualization techniques and guidelines, particularly those aligned with Gestalt Principles of Visual Perception. To demonstrate an empirical understanding of data visualization, I select a specific graphic from the paper and enhance it by applying advanced visualization techniques. I then offer a comprehensive explanation of my design choices, detailing why my revised graphic more effectively conveys information compared to the

original. The report concludes with an evaluation of the authors' work and their communication approach to their audience, accompanied by specific suggestions for improving the paper's graphical presentations.

2. An Examination of Visualisation Techniques

In the chosen article, 13 visual elements were used to present the results of a systematic literature review (SLR) evaluating Twitter's predictive potential. The findings were predominantly presented using tables, with the exception of two figures generated by an LDA machine learning algorithm and only a line graph created from Journal Citation Reports (JCR) database search results. Notably, there was limited effort to graphically represent the evaluation results, which could have enhanced the audience's understanding.

The Methodology section detailed the authors' comprehensive database searches across six data sources: ScienceDirect, Scopus, Web of Science, IEEE, Google Scholar, and Dimensions.ai. Their search query specifically targeted Twitter-related predictive or forecasting studies. Table 1 effectively summarizes the search results, presenting the number of publications found in each data source. In this context, a horizontal bar chart would be a more effective visualization to compare the number of discoveries across different data sources. This chart or graph type would allow for easy comparison of the data values while accommodating potentially long database names. While tables engage our verbal processing system, graphs engage our visual processing system, which is more efficient at quickly interpreting information (Knaflitz, 2015, p.43).

Table 1
Distribution of publications per source.

Sources	No. of papers
Scopus	1,739
Dimensions.ai	2,271
IEEE	914
Web of Science (WoS)	1,000
Science Direct	2,004
Total	7,928

Table 2
Inclusion criteria for the articles in the scope.

Phase	Criteria	Publications in scope
Database creation	Match search query	7,928
Search results	No duplicates	4,465
	No valid DOI or lack of DOI	
Exclusion of descriptive articles	Not applicable	3,250

Table 2 outlines the process of refining the search results to obtain relevant publications. The initial collection yielded 7,928 publications. The refinement process involved several steps: removing duplicates and invalid DOIs, filtering out papers unrelated to the research scope, excluding non-English publications, and eliminating research that did not use Twitter as an analytical or predictive system. This systematic approach resulted in a final selection of 3,250 contributions. While the table effectively summarizes these steps, a waterfall chart or Excel's process diagram could have provided a more visually intuitive representation of this data cleaning process, allowing readers to quickly grasp the reduction in publication numbers at each stage.

Table 3 presents the core findings of the research work, categorizing 3,250 articles published between 2009 and 2021 into 15 distinct application domains. This comprehensive breakdown

provides a crucial overview of the research landscape across different areas of Twitter-based predictive studies.

Table 3
Distribution of publications per Year and Application Domain.

Application Domain	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Total
Sentiment analysis		2	6	27	34	49	63	82	73	109	97	153	87	782
Communities on Social Media		4	8	15	38	51	57	65	76	79	58	94	75	620
Healthcare & Public Health	1		4	4	11	13	27	30	51	59	48	95	133	476
Sales & Stock Prediction	1	1	4	5	9	9	22	37	37	39	50	60	54	328
Society	1		4	9	16	16	16	31	30	36	32	49	31	271
Politics	1	1	3	7	13	19	19	19	33	21	32	38	29	235
Disaster Management		1	2	3	5	10	12	14	11	19	22	31	30	160
Location Estimation		1	2	5	7	11	10	16	16	25	14	22	8	137
Business		1	1	1	4	4	4	7	9	9	13	15	9	77
Transportation						1	3	7	9	6	15	6	10	57
Bibliometrics			1			1	1	2	6	3	6	8	7	35
Others					2	1	2	3	1	3	5	6	2	25
Demographics					1		4	4	1	2	4	1	3	20
Sports						1	3	1	5	3	1	2	1	17
Language				1		1		2	1	1	1	3		10
Total	4	11	35	77	140	187	243	320	359	414	398	583	479	3,250

While the table employs a colour heatmap to enhance readability, the abundance of numerical data can be time-consuming to interpret. To offer a more visually appealing and easily comprehensible overview, a multi-line chart or stacked area chart could be more effective, as these formats excel at illustrating changes in publication numbers over time. For the purpose of this report, Table 3 has been selected as the focus for enhancing the graphical representation of the data. The specific alterations and their rationale will be elaborated upon in the subsequent section.

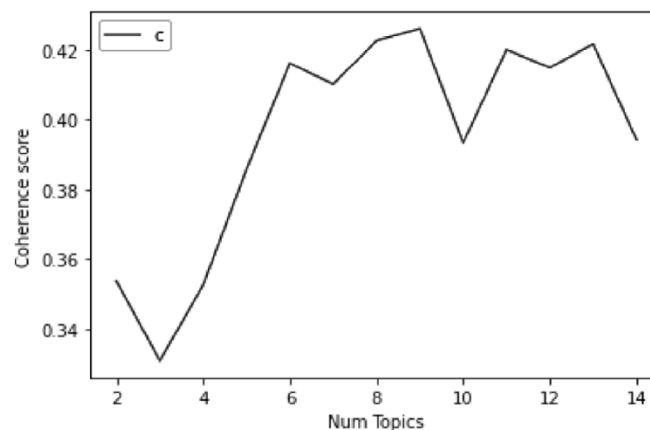


Fig. 1. C_v coherence evolution per number of topics in lda algorithm.

In the Results section, the authors present a line graph (Fig. 1) generated by the Latent Dirichlet Allocation (LDA) algorithm. This graph is used to assess the accuracy of topic classification (also referred to as application domains) by plotting the coherence scores of topics. While this visualization is derived from a sophisticated algorithm, it lacks detailed explanation that would make it accessible to a general audience. The technical nature of the LDA algorithm and coherence scores necessitates a more comprehensive explanation to ensure that readers without specialized knowledge can fully grasp the significance of the results presented in the graph.

Table 4
Latent Dirichlet Allocation (LDA) inferred topics.

R	Inferred Topic	Keywords
1	Twitter as a predictive system	user, network, model, information, predict
2	Location estimation	user, location, time, predict, event
3	Sentiment analysis, popularity and anomaly prediction	sentiment, analysis, feature, language, text
4	Society	model, sentiment, prediction, learning, personality
5	Community on Social Media	use, communication, message, engagement, community
6	Sales & Stock prediction	stock, market, sentiment, predict, price
7	Healthcare & Public Health	Covid, sentiment, public, analysis, health
8	Politics	election, political, predict, result, candidate
9	Bibliometrics	article, impact, citation, Altmetrics, journal

Table 4 presents nine inferred topics and their associated keywords generated by the LDA algorithm. These topics align with the previously identified application domains, demonstrating the algorithm's effectiveness in capturing the main themes of the research. The tabular format effectively displays this information, as it allows for a clear comparison between the algorithmically derived topics and the manually labelled application domains. Therefore, the current tabular presentation is appropriate and does not require further visualization.

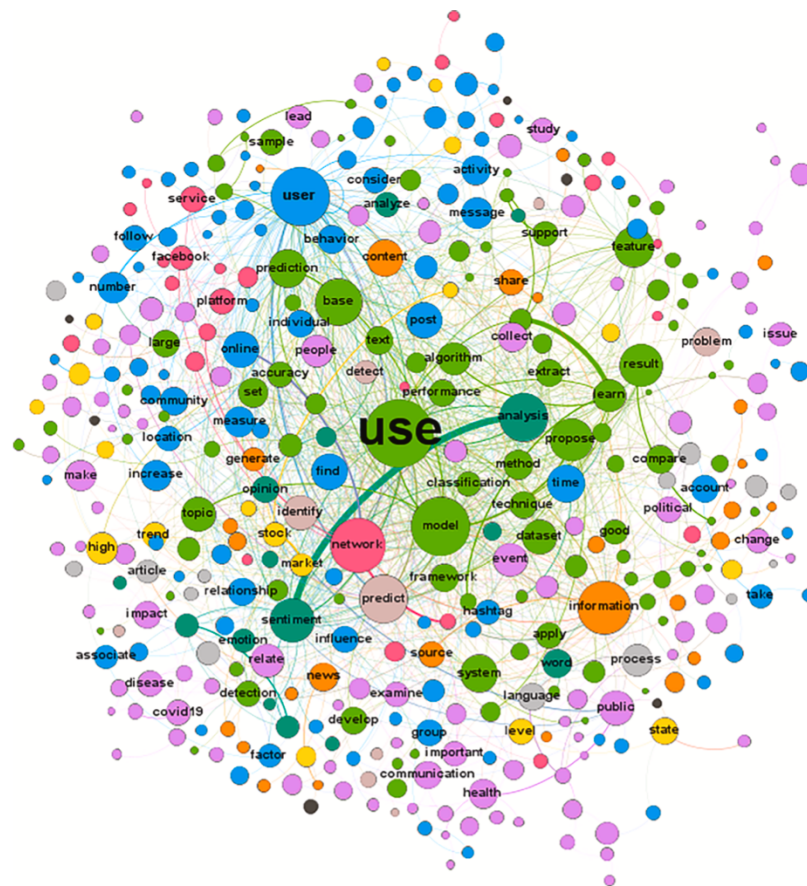


Fig. 2. Abstract & Titles Text-to-Network analysis for topic identification.

Fig. 2 presents a word cloud generated by a text-to-network analysis algorithm, using the abstracts and titles of the articles as input. This machine-generated image provides a visual representation of the most influential keywords in the corpus. Complementing this visualization, Table 5 quantifies the weight of inferred topics as percentages of the 3,250 articles analysed.

The word cloud offers an intuitive, at-a-glance view of prominent themes, while Table 5 provides a more precise breakdown of topic distribution. Together, these elements offer a comprehensive understanding of the thematic landscape of the research. The word cloud serves as a visually engaging introduction to the key concepts, which are then elaborated upon in the tabular format of Table 5. Given that these two elements effectively complement each other, additional visual support for this particular aspect of the analysis is not necessary.

Table 5
Text-to-network analysis of articles network's inferred topics.

R	C	Weight (%)	Inferred Topic	Keywords
1		29.87	Twitter as a predictive system	use, prediction, dataset, algorithm, learn
2		16.79	Community on Social Media	user, activity, behaviour, community, relationship
3		11.55	Healthcare & Public Health	public, health, impact, disease, covid19
4		9.01	Twitter as real-time data source Sales & Stocks prediction	source, knowledge, real-time, stream, financial, insight
5		8.22	Sentiment analysis, popularity and anomaly prediction	analysis, sentiment, opinion, investor, polarity
6		6.90	Business Twitter platform	network, platform, product, review, microblog
7		5.71	Sales & Stocks prediction	trend, stock, market, company, return, price
8		5.49	Twitter as a predictive system in Public Health	predict, identify, problem, detect, analyse, depression

Table 6 presents a detailed breakdown of sub-topics derived from the main inferred topics, along with their respective weights calculated as percentages of the analysed articles. This table also maps these inferred sub-topics to the previously established classification of application domains. The multi-dimensional nature of this data - encompassing main topics, sub-topics, their weights, and corresponding application domains - is effectively captured in the tabular format. This presentation allows for a comprehensive view of the thematic structure within the research corpus, enabling readers to understand the nuanced relationships between broader topics and their more specific sub-components.

Table 6
Inferred topics by application domain.

Application domain	Weight (%)	Inferred Topic	Keywords
Sentiment analysis, popularity and anomaly prediction	17.26	Text analysis	feature, process, text, classification, language
	16.38	Predictive techniques	model, prediction, learn, algorithm, machine
	15.34	Popularity prediction	user, content, online, post, influence, retweet
	10.64	Sentiment analysis	sentiment, analysis, opinion, public, emotion
Community on Social Media	16.61	Community detection techniques	model, algorithm, cluster, graph, classification
	15.22	Prediction on communities	user, predict, node, group, pattern
	12.53	Community detection	community, word, people, analyze, hashtag
	12.25	Social Network analysis	network, influence, structure, dynamic, spread
Healthcare & Public Health	21.21	Mental health	people, individual, patient, population, depression
	14.20	Twitter as predictive system in healthcare	model, predict, learn, method, compare
	11.11	Disease outbreak prediction and analysis	health, pandemic, response, impact, help
	9.10	Pandemic	outbreak, influenza, surveillance, spread, virus
Sales & Stock Prediction	23.23	Stock price prediction	stock, market, predict, price, company
	16.07	Financial forecasting	model, analysis, forecast, performance, compare
	14.60	Cryptocurrencies	factor, cryptocurrencies, event, technology, platform
	13.50	Investment & Sales	sentiment, investor, opinion, public, identify
Society	18.67	Twitter as social data source	use, post, people, word, analyse
	18.50	Study of the individual	predict, personality, behaviour, influence, individual
	7.88	Crime prediction	crime, prediction, increase, news, investigate
Politics	31.15	Election results prediction	election, prediction, politician, electoral, presidential
	21.53	Political communities	political, sentiment, opinion, people, trend
	9.02	Political analysis	candidate, party, vote, lead, speech
Energy, Environment and Natural Disaster Management	10.09	Disaster Management	emergency, communication, response, crisis, fire
	9.30	Disaster prediction	disaster, human, damage, natural, critical
	7.90	Disaster identification Weather forecast	flood, weather, risk, forecast, impact, water
Location Estimation	23.81	User geolocation	location, user, post, content, geographic
	14.25	Prediction on Tourism	use, collect, tourist, mobile, destination
	12.00	Community geo-detection	prediction, method, cluster, geolocation, accuracy

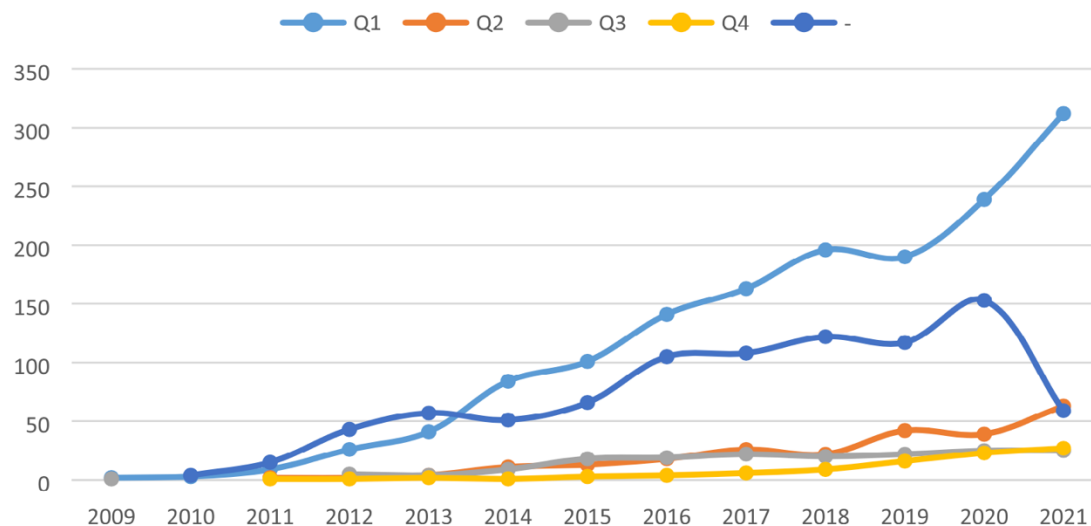


Fig. 3. Evolution of the number of publications in the different quartiles (JCR).

Fig. 3 presents the number of publications of Table 7, which measures the impact factors of the most relevant journal articles about Twitter. These data are sourced from the Journal Citation Reports (JCR) database. However, there are several issues with the Fig. 3 that hinder clear interpretation:

1. The connection between Fig. 3 and Table 7 is not explicitly established, which may confuse the audience while looking at the graph.
2. The y-axis in Fig. 3 lacks labels, making it difficult to understand the scale of the impact factors.
3. The legend does not clearly explain that "Q1" refers to "Expert Systems with Applications," the top-ranked journal in the dataset.
4. The line graph in Fig. 3 could benefit from significant enhancements to better utilize the data presented in Table 7.

Table 7
Most relevant journals.

Journal	Times Cited	Number of Publications	Impact factor
Expert Systems with Applications	2,864	83	6.954
Computers in Human Behavior	4,443	79	6.829
Information Processing and Management	1,270	56	6.222
Knowledge-Based Systems	1,095	40	8.038
PLoS ONE	2,033	40	3.240

The remaining tables—Table 8 (Most Relevant Conferences Series), Table 9 (Top 20 Most Cited WoS Articles), and Table 10 (Top 5 Articles with the Highest Impact Potential)—are also sourced from the JCR database. The data presented in these tables are organized in a manner that enhances their appropriateness for analysis and interpretation.

Table 8
Most relevant conferences series.

Conference	Times Cited	Number of Publications
Procedia Computer Science	1,248	110
ACM International Conference Proceeding Series	384	49
International Conference on Information and Knowledge Management	958	25
Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining	779	13

Table 9
Top-20 Most cited (WoS) articles.

ID	References	Application domain	Topic	Aim	Times cited (WoS)
S1	(Bollen et al., 2011)	Sales & Stock Prediction	Stock price prediction	Mood analysis in Twitter to predict the stock market with neural networks	2,615
S2	(Kalchbrenner et al., 2014)	Language	Social Network analysis	Language understanding for semantic modelling of sentences	1,393
S3	(Asur & Huberman, 2010)	Sentiment Analysis, popularity & anomaly prediction	Social Network Analysis	Twitter as a predictive system in the real-world	1,101
S4	(Suh et al., 2010)	Sentiment Analysis, popularity & anomaly prediction	Popularity prediction	Aggregated Twitter data to predict retweetability with content features	818
S5	(Signorini et al., 2011)	Healthcare & Public Health	Disease outbreak prediction and analysis	Twitter as a measure of public interest about health	768
S6	(Eysenbach, 2011)	Bibliometrics	Citation prediction	Usefulness to predict the citations of an article based on social media activity	658
S7	(Eysenbach, 2009)	Healthcare & Public Health	Public Health	Twitter activity to predict public health infodemiology	610
S8	(Hughes et al., 2012)	Community on Social Media	Community detection	Relationship between personality and the use of social networking sites	572
S9	(Colleoni et al., 2014)	Politics	Political communities	Predict Political orientation through network analysis	524
S10	(He et al., 2013)	Business	Market analysis	Social media competitive analysis to show the value of text mining	519
S11	(Cheng et al., 2014)	Community on Social Media	Social Network Analysis	Predict content cascades in social networking websites	482
S12	(Smith et al., 2012)	Business	Market analysis	General framework for comparing brand-related user content	479
S13	(Hong et al., 2011)	Sentiment Analysis, popularity & anomaly prediction	Popularity prediction	Twitter stats to classify and predict the popularity of tweets	421
S14	(Schultz et al., 2011)	Community on Social Media	Social Network Analysis	Twitter as medium for crisis communication	419
S15	(Golbeck et al., 2011)	Society	Study of the individual	Personality prediction based on Twitter stats and text analysis	379
S16	(Gerber, 2014)	Society	Social unrest and crime	Crime prediction system based on public data from Twitter	360
S17	(Takhetev et al., 2012)	Location Estimation	Social Network Analysis	Geographic analysis of Twitter network to predict ties between people and places	348
S18	(Achrekar et al., 2011)	Healthcare & Public Health	Disease outbreak prediction and analysis	Seasonal flu epidemics evolution based on public data	326
S19	(Saif et al., 2012)	Sentiment Analysis, popularity & anomaly prediction	Sentiment analysis	Sentiment analysis applied to Twitter data to monitor public opinion	279
S20	(Quercia et al., 2011)	Society	Study of the individual	Personality prediction based on popularity	317

Table 10
Top-5 articles with the highest impact potential.

ID	References	Application domain	Topic	Aim	Times cited (WoS)	C _n
	(Bollen et al., 2011)	Sales & Stock Prediction	Stock price prediction	Mood analysis in Twitter to predict the stock market with neural networks	2,615	237.7
S1	(Onan, 2021)	Sales & Stock Prediction	Investment & Sales	Deep learning techniques applied to product reviews and sentiment analysis	45	207.9
S2	(Liu et al., 2021)	Sales & Stock Prediction	Investment & Sales	Customer engagement analysis with NLP applied to UGC	60	60.3
S3	(Oliveira et al., 2017)	Sales & Stock Prediction	Stock price prediction	Twitter usefulness to forecast stock market variables	152	31.5
S4	(Shen et al., 2019)	Sales & Stock Prediction	Cryptocurrency	Twitter activity as a reflect of trading volume of Bitcoin	94	29.8
S5	(Bartov et al., 2018)	Sales & Stock Prediction	Stock price prediction	Aggregate opinion to assess to future value of a company stock	100	26.1

3. Data Visualisation and Interpretation

The chosen graph for enhancement presents the distribution of publications per year and their application domain areas using a heatmap table with precise statistical figures. Knafllic (2015, p. 42) notes that heatmaps facilitate data analysis by using colour saturation to visually represent numerical values, allowing readers to quickly identify areas of interest, such as extreme values in a dataset.

However, Table 3 (refer the section above), which lists application domains and their publication counts from 2009 to 2021, presents challenges in quickly grasping the data's narrative. A multiline chart or area chart could more effectively and rapidly convey the story within the data, particularly as it evolves over time. This approach is especially relevant given the readers' interest in the evolution of publications in Twitter's most researched areas as a predictive system.

In creating a new multiline graph, application domain names can be represented in the legend, with years on the x-axis. Following Knafllic's (2015) guidelines, I prepared the data in Excel from Table 3 and chose a multiline graph with vibrant colours in Microsoft Excel to illustrate trends and variations over time.

To enhance the data visualization, I implemented the following steps:

1. Set the chart title as "Distribution of Publications per Year & Application Domains (2009-2021)" for clarity.
2. Added data labels to provide detailed figures aligned with years, adhering to the Gestalt principle of proximity (Knafllic, 2015, p. 75).
3. Removed gridlines and the primary vertical axis to reduce cognitive load and eliminate clutter (Knafllic, 2015, p. 71-74).
4. Included the horizontal axis title "Year" for clarity.
5. Moved the legend to the right, placing it close to the lines to allow readers to associate the colours of "Application Domains" with trend lines, following the Gestalt principle of similarity (Knafllic, 2015, p. 76).

Ajani *et al.* (2022) has provided two key guidelines for effective data visualization; first is to declutter the visual elements and second is to focus on relevant data. Therefore, these modifications aim to create a more effective and visually appealing representation of the data, allowing readers to quickly grasp trends in publication distribution across different application domains over time.

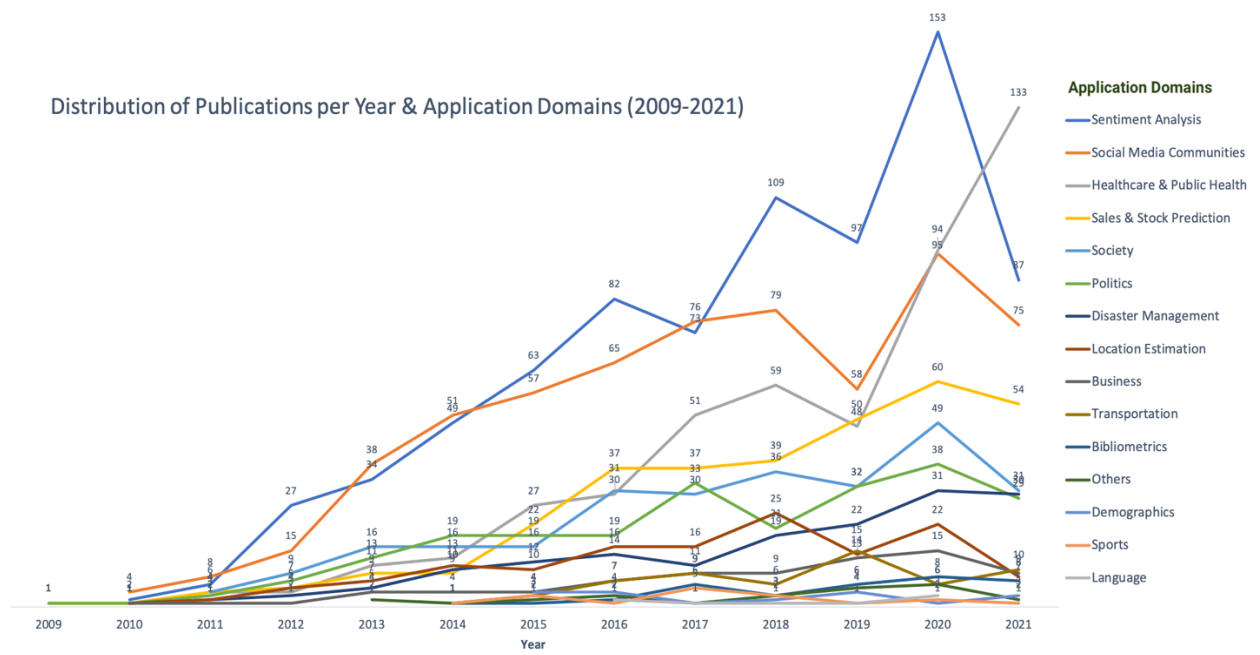


Fig. 4 Table 3 Data Transformed to Multiline Graph using Microsoft Excel

The study examines research and business applications derived from Twitter data analysis and its predictive potential. The authors categorized Twitter-related research, with the results presented in Table 3 and now visualized in an enhanced multiline chart. The enhanced diagram reveals several key trends in publication counts across different application areas:

- Sentiment Analysis emerges as the most prominent category, underscoring its significance in Twitter analytics. The trend line shows a consistent increase from 2010 to 2020, peaking at 153 publications, before experiencing a sharp decline to 87 publications in 2021.
- Communities on Social Media ranks as the second largest application area. Publications in this category show an upward trend from 2010 to 2018, followed by a dip in 2019. The numbers then fluctuate, rising to 94 in 2020 before slightly decreasing to 75 in 2021.
- Healthcare & Public Health demonstrates a remarkable growth trajectory, particularly in recent years. This category, which includes research on outbreak detection, epidemic/pandemic evolution, and public health management, shows a steady increase before 2020 and a dramatic surge thereafter, reaching 133 publications in 2021 - likely influenced by the COVID-19 pandemic.

Other notable categories include:

- Sales & Stock Prediction: Shows consistent growth, reaching 54 publications in 2021.
- Society: Experiences fluctuations, with a peak of 49 publications in 2020.
- Politics: Demonstrates variable interest, with 38 publications in 2020 and 29 in 2021.
- Disaster Management: Shows a gradual increase, reaching 30 publications in 2021.
- Location Estimation: Peaks at 25 publications in 2018 before declining to 8 in 2021.

Less prominent but still significant categories include Business, Transportation, Bibliometrics, Demographics, Sports, and Language processing. These areas show varying patterns of growth and decline over the years.

The "Others" category, encompassing uncategorized publications such as cyberattack identification and AI impact analysis in workplaces, shows modest numbers throughout the period.

This comprehensive overview illustrates Twitter's diverse applications as a predictive system, offering valuable insights for researchers and business leaders across multiple domains. The data reveals how research interests have evolved over time, with some areas showing consistent growth while others display more variable patterns, often reflecting broader societal and technological trends.

4. Conclusion

This data visualization analysis of SLRs on Twitter as a predictive system has revealed several significant trends and insights across various application domains from 2009 to 2021.

The enhanced multiline chart effectively illustrates the evolution of research interests in Twitter analytics over time, providing a clearer and more intuitive representation of the data compared to the original heatmap table. This visualization allows for quick identification of trends and comparisons between different application domains.

Key findings from the visualization include:

1. Sentiment Analysis emerged as the dominant research area, showing consistent growth until 2020 before experiencing a slight decline in 2021. This trend underscores the continued importance of sentiment analysis in Twitter-based predictive systems.
2. Communities on Social Media maintained its position as the second most researched area, demonstrating the ongoing interest in social network analysis techniques applied to Twitter data.
3. Healthcare & Public Health exhibited remarkable growth, particularly in 2020 and 2021, likely influenced by the COVID-19 pandemic. This surge highlights Twitter's increasing relevance in public health monitoring and management.
4. Other areas such as Sales & Stock Prediction, Society, and Politics showed varying patterns of growth and fluctuation, reflecting the diverse applications of Twitter data in predictive systems.
5. The visualization revealed the impact of external events, such as the COVID-19 pandemic, on research trends, particularly evident in the Healthcare & Public Health category.

The transition from a heatmap table to a multiline chart has significantly enhanced the data's storytelling capacity. It allows for easier identification of long-term trends, sudden changes, and comparisons between different application domains. This approach aligns with best practices in data visualization, as outlined by Knaflie (2015), by reducing cognitive load and emphasizing key information.

In conclusion, this visualization analysis provides a comprehensive overview of the evolving landscape of Twitter-based predictive systems research. It highlights the diverse applications of Twitter data across multiple domains and demonstrates the platform's growing importance as a tool for prediction and analysis in various fields. The insights gained from this visualization can guide researchers and practitioners in understanding the current state and future directions of Twitter analytics research.

5. References

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6. Appendix

The multiline chart was initially generated using Python's pandas and matplotlib libraries as part of my personal exploration of the data. This visualization served as a starting point for understanding the trends in different application domains over time. Building upon this initial chart, I further enhanced and refined the visualization using Microsoft Excel. This iterative process allowed me to leverage the strengths of both Python's data manipulation capabilities and Excel's user-friendly interface for fine-tuning the visual presentation. The resulting chart provides a clear and comprehensive view of the publication trends across various application domains from 2009 to 2021, based on the data provided in the CSV file.

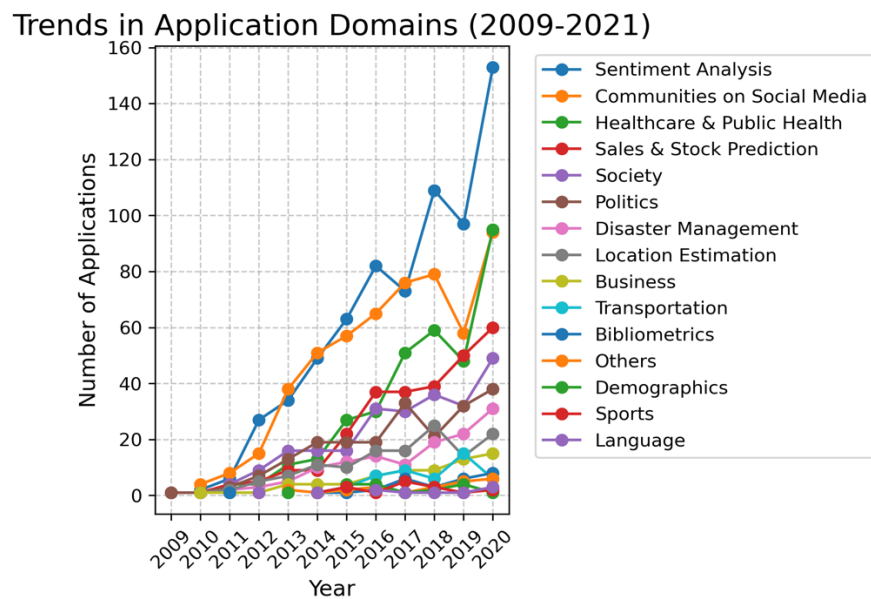


Fig. 5. Table 3 Data Transformed to Multiline Graph using Python

Python Source Code

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the CSV file includes data of Table 3
df = pd.read_csv('RevisedGraph.csv')

# Set the 'Application Domain' column as the index
df.set_index('Application Domain', inplace=True)

# Transpose the dataframe to have years as index
df_transposed = df.transpose()

# Create the line plot
plt.figure(figsize=(21, 16))

# Plot lines for each application domain
for column in df_transposed.columns:
    plt.plot(df_transposed.index[:-1], df_transposed[column][:-1],
            marker='o', label=column)

# Customize the plot
```

```
plt.title('Trends in Application Domains (2009-2021)', fontsize=16)
plt.xlabel('Year', fontsize=12)
plt.ylabel('Number of Applications', fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', fontsize=10)
plt.grid(True, linestyle='--', alpha=0.7)

    # Rotate x-axis labels for better readability
plt.xticks(rotation=45)

    # Adjust layout to prevent cutting off labels
plt.tight_layout()

    # Save the plot as a high-resolution PNG file
plt.savefig('application_domains_trends.png', dpi=300, bbox_inches='tight')

    # Display the plot
plt.show()
```